Mammogram Analysis Challenge

Anisha, Hongyu, Sofie

Problem

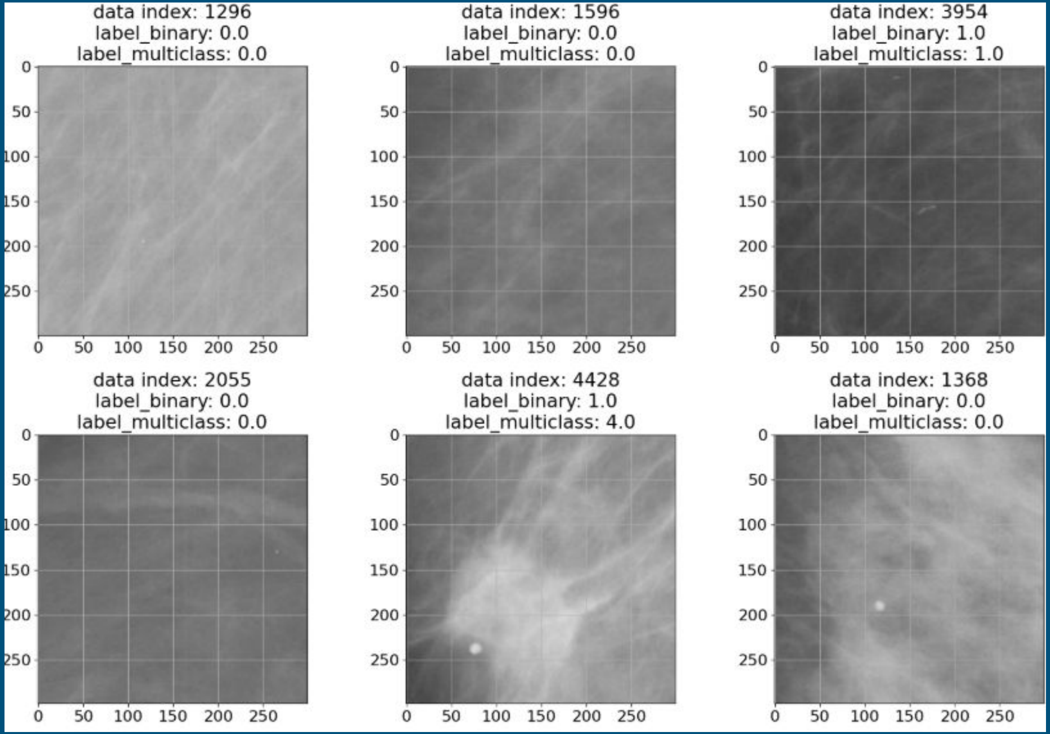
- Breast cancer is one of the most common cancers among women worldwide
- Early detection of breast cancer through mammograms significantly increase survival rates
- **Mammograms can be difficult to interpret and variability in image quality/human interpretation leads to many false positives and negatives**

Impact

We aim to train smart model to better assist doctor in detecting breast cancer. Utilizing advanced model will help to avoid false negative and false positive in repeated medical consultation.

Dataset/Visualization

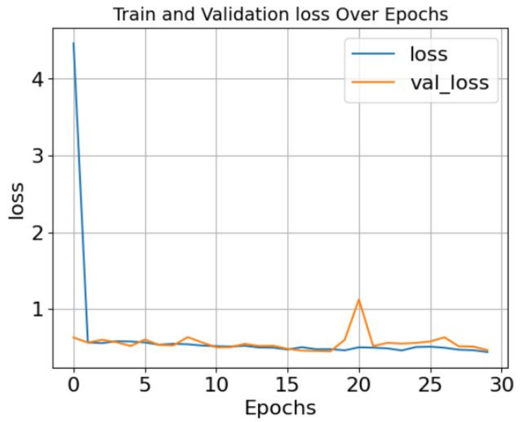
- Mammogram segments
- Binary labels - positive for abnormality
- Multiclass labels - negative, positive benign calcification, positive malignant calcification, positive benign mass, and positive malignant mass



Binary Classification

- Baseline model was just a series of convolutional layers
- Implemented a vision transformer + manual search for optimal parameters
  - Learning rate scheduler (Exponential Decay and Cosine)
  - Weight decay using AdamW optimizer
  - Dropout = 0.2
  - Transformer Encoder Layers = 16
  - Data augmentation: rotation, flips, etc.
- ConvNextV2 Architecture
  - Ideal for medical imaging tasks where data may be limited or noisy
  - Learning rate scheduler

Binary Classification Results



Initial Accuracy with baseline model:

- 0.69 on training set, 0.48 on validation set

learning rate schedule, and optimized parameters

- 0.8061 on training set, 0.775 on validation set, 0.83 on mock test set

Vision Transformer Model

- 0.85 on training set, 0.8425 on validation set, 0.84 on mock test set

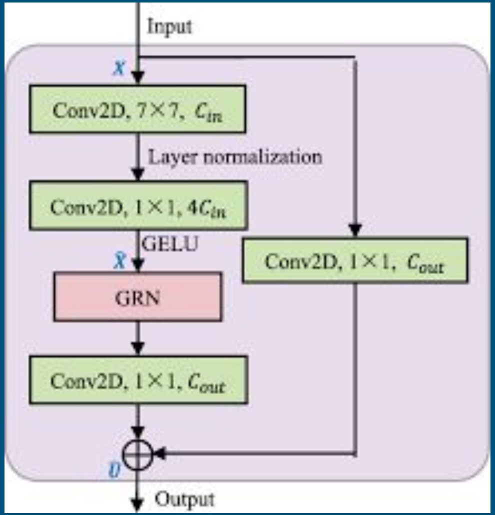
ConvNextV2

- 0.77 on validation set

AUROC = 0.81
Best Accuracy = 0.8425

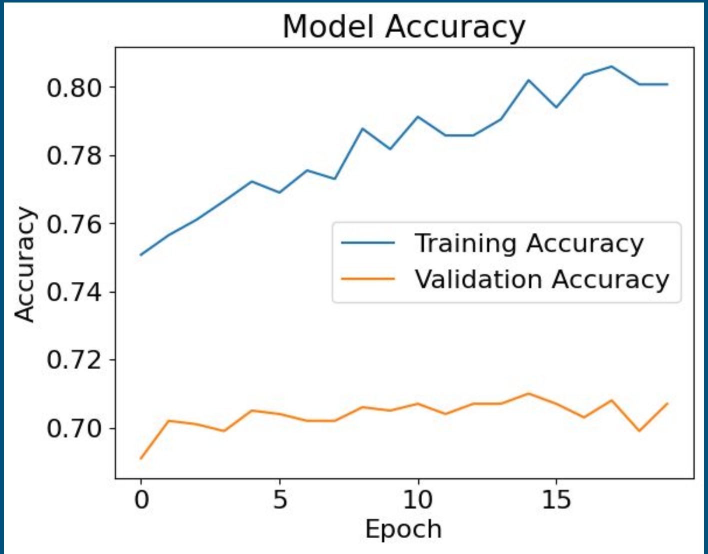
Multiclass Classification

- Learning rate scheduler
    - Optimized other parameters (dropout, number of layers etc)
- Vision Transformer
    - Changed code for binary for multiclass (five classes)
- Transfer Learning
    - VGG16 model
- Convnext V2
    - Resize images to 224 224
    - 3 channels (initially gray scale)

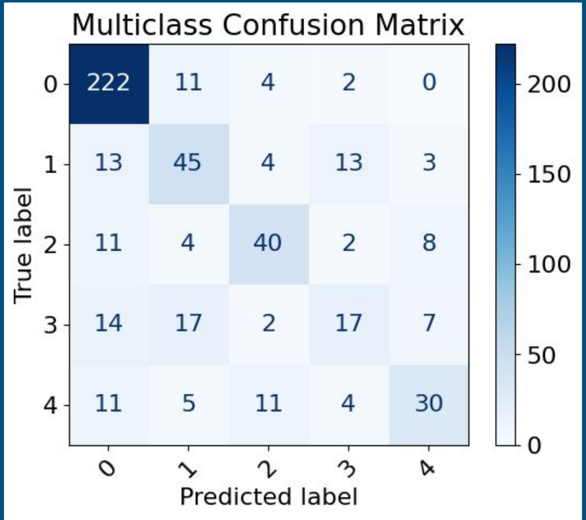


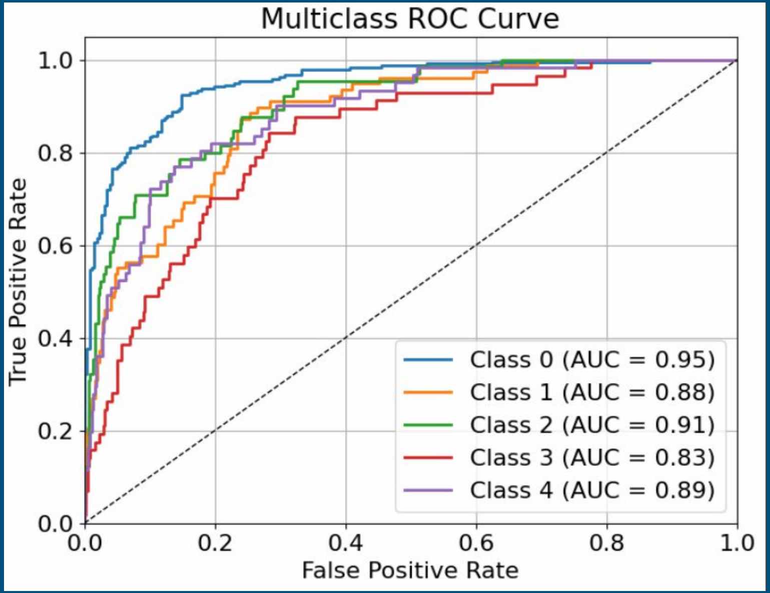
Multiclass Classification Results

- ConvNextV2 gave the best results
  - Initially, before tuning parameters: 0.654 accuracy on validation set
  - After adding customized layers (global average pooling, batch norm, dense layers and dropout): 0.682 on validation set
  - After tuning it more and adding a learning reate scheduler: 0.71 on validation set
- Transfer learning + learning rate scheduler
  - 60% accuracy on validation set



Results - Multiclass





Challenges

- Limited computational power
    - Requires smaller but accurate model
    - Tried to split models and test it in different sections
    - Gained more ideas in model designing
- Changing dimensions and color of image for transfer learning compatibility
    - Images were in gray-scale, duplicated last dimension for it to have three channels
- Improving validation accuracy
    - Initially, the validation accuracy ranges from 50% - 70%, requires trials and error in learning rate, number of layers and batch size
    - Used data sheet to record trials
    - Improved in model testing

Future Improvement & Expansion

Improvement

- Grid Search to optimize hyperparameters of model

Expansion

- Expand data size through collaboration with medical centers
- Ensemble model